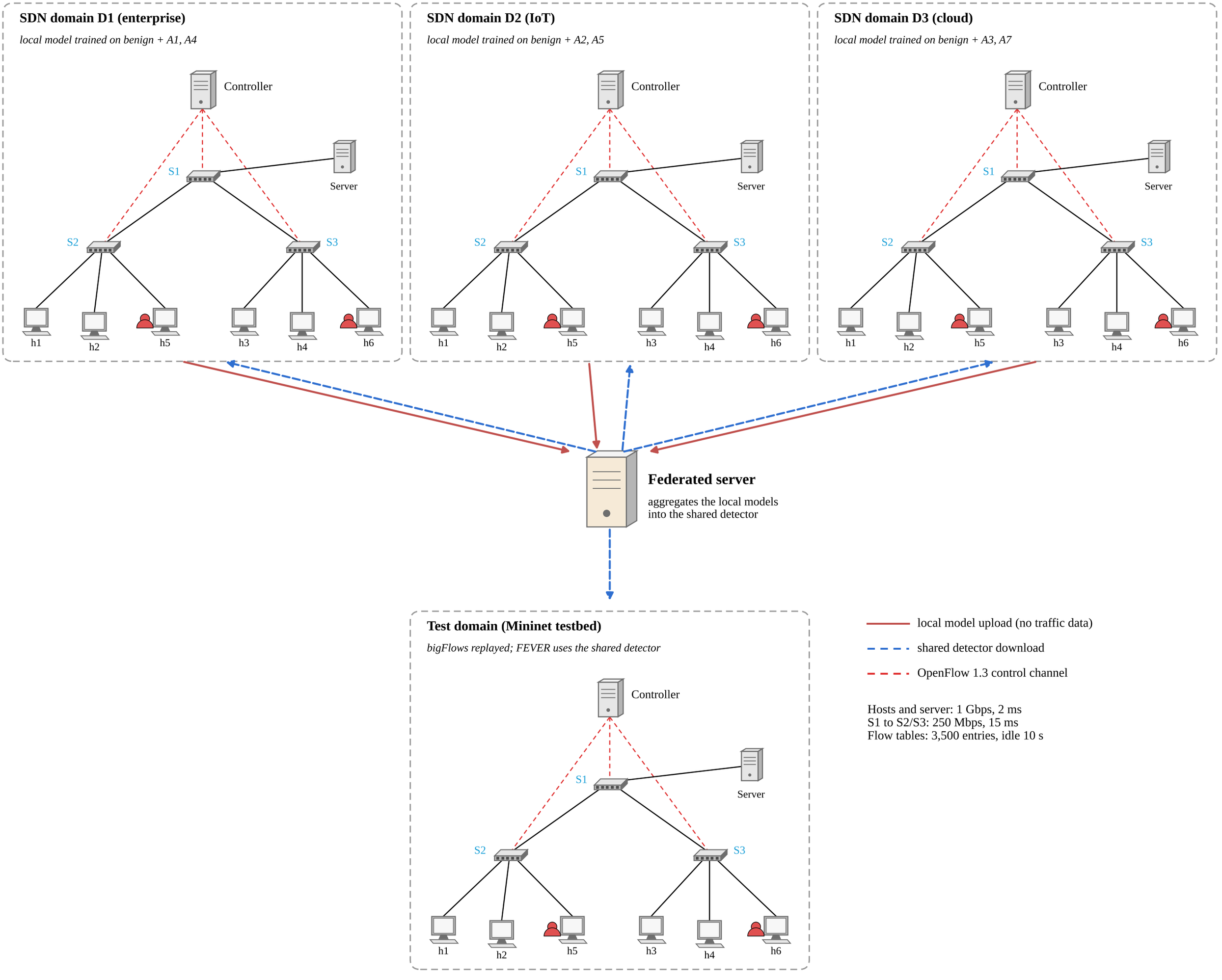
SDN domain D1 (enterprise)
local model trained on benign + A1, A4
Controller
S1
Server
S2
S3
h1
h2
h5
h3
h4
h6
SDN domain D2 (IoT)
local model trained on benign + A2, A5
Controller
S1
Server
S2
S3
h1
h2
h5
h3
h4
h6
SDN domain D3 (cloud)
local model trained on benign + A3, A7
Controller
S1
Server
S2
S3
h1
h2
h5
h3
h4
h6
Federated server
aggregates the local models into the shared detector
Test domain (Mininet testbed)
bigFlows replayed; FEVER uses the shared detector
Controller
S1
Server
S2
S3
h1
h2
h5
h3
h4
h6
local model upload (no traffic data)
shared detector download
OpenFlow 1.3 control channel
Hosts and server: 1 Gbps, 2 ms
S1 to S2/S3: 250 Mbps, 15 ms
Flow tables: 3,500 entries, idle 10 s